Storm Sequence Generator — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 02-April-2026

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1 Test Results — Storm Sequence Generator (MKM-SS-001)

Tests run on **02 April 2026 at 18:34:14**.

Table 1: Test Results Summary — Storm Sequence Generator (MKM-SS-001)

Total Tests	17
Passed	17
Failed	0
Skipped	0
Duration	0.00s

Table 2: Detailed Test Results — Storm Sequence Generator

Test Description	**Acceptance Criteria**	**Result**	**Time**
Duration must not exceed MAX_PRECIP_END_HOUR (156).	`result.storms[0].duration_hours <= MAX_PRECIP_END_HOUR`	Pass	0.000s
Isolated has one storm	`isinstance(result, StormSequence); result.num_storms == 1` (+2 more)	Pass	0.000s
Aggregate stats	`result.max_intensity_factor >= result.avg_intensity_factor; result.total_precipitation_mm > 0` (+1 more)	Pass	0.000s
Catchment id in storms	`result.storms[0].catchment_id == "rhine"`	Pass	0.000s
Cluster sequence	`result.sequence_type in (SequenceType.CLUSTER.value, Sequ...`	Pass	0.000s
force_sequence_type='doublet' → always doublet.	`result.sequence_type == SequenceType.DOUBLET.value; result.num_storms == 2`	Pass	0.000s
force_sequence_type='isolated' → always isolated.	`result.sequence_type == SequenceType.ISOLATED.value; result.num_storms == 1`	Pass	0.000s
Minimal intensity (5% sequence probability) mostly produces isolated.	`isolated >= 10`	Pass	0.001s
Generate multiple sequences without error.	`isinstance(result, StormSequence); result.num_storms >= 1`	Pass	0.000s
Persistent sequence	`result.sequence_type in (SequenceType.PERSISTENT.value, S...`	Pass	0.000s
Returns storm sequence	`isinstance(result, StormSequence)`	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
All sequences must have total_duration ¡= 168h.	`result.total_duration_hours <= 168.0`	Pass	0.000s
Storms have correct fields	`storm.duration_hours > 0; storm.intensity_factor > 0` (+2 more)	Pass	0.000s
Custom catchment	`gen.catchment_id == "rhine"; gen.base_precipitation == CATCHMENT_BASE_PRECIP["rhine"]`	Pass	0.000s
Default catchment	`gen.catchment_id == "thames"; gen.base_precipitation == CATCHMENT_BASE_PRECIP["thames"]`	Pass	0.000s
Same seed → same sequence.	`seq1.num_storms == seq2.num_storms; seq1.sequence_type == seq2.sequence_type`	Pass	0.000s
Unknown catchment defaults to 35	`gen.base_precipitation == 35.0`	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
12-Mar-2026	1.0	Initial test suite (116 tests)	D. Kelly
27-Mar-2026	1.1	64 test(s) removed (total: 52)	D. Kelly
02-Apr-2026	1.2	35 test(s) removed (total: 17)	D. Kelly